

Course Project Checkpoint 1

Rotating Pendulum Stability Sensor Platform

1. Project Idea

My project is a battery-powered rotating pendulum stability sensor platform. The system is made of a circular base with a standing pendulum arm attached near the outside. A small weight at the tip of the arm makes the system have more rotational inertia when upright before falling. The aim here is to detect the orientation and position of the pendulum by collecting information from different sensors, processing this information to obtain a stability factor. The system will incorporate a motor to rotate the base, and thus keeping it upright.

This idea came from watching videos on the three-body problem about a year ago and seeing someone pull off a fully controlled double pendulum, which stuck with me. After talking to Dr. Trudgen I figured this was the right class to actually build something like it, and honestly it would be cool to show my younger cousins and maybe get them interested in engineering. The project connects to real systems like balancing robots and camera gimbals, but for now I am focusing on measuring other things like tilt and position, and conditioning the raw sensor signal, and also fusing the data with a Kalman filter, while logging everything over Wi-Fi.

2. Project Constraints

For this project, the RAW sensor being used is a Hall effect analog sensor like the A1302 coupled with a tiny neodymium magnet taped on the end of the pendulum. When the pendulum tilts, the magnetic flux varies due to the magnet getting nearer or farther from the sensor, producing an output voltage signal that will change in a non-linear fashion since the magnitude of the magnetic field inversely varies with distance raised to the third power. The raw voltage signal produced is what I am going to use. After the active amplification using the op-amp, the signal will then be filtered using a low-pass filter to remove any high frequency noise in the signal prior to feeding it to the ADC pins of the ESP32. The Linearizer circuit will be done in CP3 and the idea will be to limit the operational range to that near vertical in order to design a circuit to linearize the signal in that range.

The two pre-packaged sensors are an IMU (MPU6050 or ICM-20948) and an INA219 current sensor, both on I2C. The IMU measures tilt angle and angular velocity. The INA219 measures motor current draw, which indicates how hard the motor is working and reflects the mechanical load on the system. Redundancy is covered because pendulum angle is estimated two independent ways: the Hall sensor uses electromagnetic field strength, while the IMU accelerometer uses gravitational mechanics. A copper tape capacitive touch pad handles user input to start, stop, or change sensitivity modes, with the ESP32 sourcing the excitation signal and reading the result. The whole system runs off an 18650 battery with a 3.3 V regulator.

3. Bill of Materials

Part	Qty.	Est. Cost	Link / Source	Notes
ESP32 development board	1	\$8–12	Adafruit #3405	Microcontroller; ADC, Wi-Fi for IoT logging
Analog Hall effect sensor (A1302 or similar)	1–2	\$2–5	DigiKey A1302EUA-T	RAW nonlinear magnetic position sensor
Small neodymium magnet	1–3	\$3–6	Amazon B00JHKFUS8	Attached to pendulum arm
MPU6050 or ICM-20948 IMU	1	\$5–15	SparkFun SEN-15335	Pre-packaged; measures tilt and angular velocity via I2C
INA219 current sensor	1	\$5–10	Adafruit #904	Pre-packaged; measures motor current draw via I2C
LM358 or rail-to-rail op-amp	1	\$1–3	DigiKey LM358AP	Active signal conditioning for RAW sensor
Resistors / capacitors (assorted)	—	~\$5	Amazon kit	Filter, gain, voltage divider, linearization network
DC gear motor	1	\$8–20	Amazon B07QXZC9TO	Rotates the base
Motor driver module (L298N)	1	\$5–10	Amazon B07BK1QL5T	Required for DC motor control
18650 battery + holder	1	\$6–12	Amazon B08FD4J2KN	Battery power source
5 V / 3.3 V regulator module	1	\$3–8	Pololu #2831	Powers microcontroller and sensors
Copper tape or copper-clad pad	1	\$3–6	Amazon B01MR5DSCM	Homemade capacitive touch input
Mechanical base materials	1 set	\$10–20	Local / 3D print	3D-printed or wood rotating base, pendulum arm, mount

Estimated total cost: approximately \$60–100; some parts are already on hand.

4. Sensor Selection

A couple options were considered for determining the pendulum's angle. The most obvious choice was a potentiometer at the axis of rotation but it suffers from added friction and wear-and-tear. An ultrasonic sensor, even though it's quite accurate it is rather bulky and slow for small deflections close to vertical orientation. While a camera can be employed for capturing the pendulum's movement, its use is unnecessary here as we need some form of a RAW sensor since it processes the image internally and outputs digital data. Finally, the digital magnetometer would react to the presence of the magnet on the pendulum arm, but the signal it would generate would no longer be RAW.

The Hall Effect sensor was the best choice, since it provides a raw voltage signal which changes in a nonlinear way with respect to the magnet distance. This signal is exactly what is needed in this class for conditioning using an operational amplifier, filtering, and linearization circuits. The IMU covers tilt and angular velocity through a completely different physical principle, giving the required redundant angle measurement from a separate branch of physics. The INA219 was chosen as the second pre-packaged sensor because the motor is part of the system, so current draw is a direct and meaningful measurement; a spike in motor current tells you the pendulum is being driven hard.

5. Standards and Testing Protocol

There is no single standard that covers this system directly, so testing will follow basic repeatability and calibration principles similar to ASTM E177. Each sensor gets zeroed against a known reference before any data is collected: Hall sensor and IMU against a level, INA219 against a known load current.

Static testing will place the pendulum at 0° , 5° , 10° , 15° , and 20° from vertical and record multiple readings from both the Hall sensor and IMU at each position to build a transfer function and compare the two methods. Dynamic testing will release the pendulum from those same angles and log the full sensor response over time. Key metrics are angular error against a reference, repeatability across trials, and whether the stability index correctly classifies stable, marginal, and unstable conditions.

6. What I Learned This Week

The main thing I took away is that the design of the project should revolve around the sensing system rather than the mechanics of it. It would be easy to approach this as a robotic balancing project and then add sensors afterward, but that is not the focus of this course. I also learned that the RAW sensor choice drives everything downstream. The Hall sensor's nonlinear response is not a problem to work around; it is actually the key feature that creates the signal-conditioning chain of amplification, filtering, and linearization required for the project.

7. AI Interaction This Week

I used AI to compare project ideas and check them against the course constraints. It helped narrow the concept down to the current sensor set but kept framing things as a control system problem rather than a sensing one, so I had to redirect it a few times. It also pushed a less relevant sensor early on, which I replaced with the INA219. Useful for organizing ideas, but the actual design calls were mine and the direction. I often find I use it more like google, helping me research and finding weak points in my ideas or understanding rather than the AI building things.